

Roll No.

TOE-801

B. TECH. (EE) (EIGHTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023
EXPERT SYSTEM AND FUZZY LOGIC

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) What is fuzzy logic and crisp logic ? Describe the fuzzy set of discrete and continuous type with examples. Also, explain membership value. (CO4, CO3)

OR

- (b) Develop a reasonable membership function for the following fuzzy sets based on height measured in centimeters : (CO3, CO4)

(i) "Tall"

(ii) "Short"

(iii) "Not short"

2. (a) Consider a simple hollow shaft of approximately 1 m radius and wall thickness $1/(2\pi)$ m. The shaft is built by stacking a ductile section, D, of

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the appropriate cross section over a brittle section, B. A downward force P and a torque T are simultaneously applied to the shaft. Because of the dimensions chosen, the nominal shear stress on any element in the shaft is T (Pascals) and the nominal vertical component of stress in the shaft is P (Pascals). We also assume that the failure properties of both B and D are not known with any certainty.

We define the fuzzy set $\underline{A}$ to be the region in (P, T) space for which material B is "safe". Similarly, we define the set $\underline{B}$ to be the region in (P, T) space for which material D is "safe" : (CO2)

$$\underline{A} = \left\{ \frac{1}{2} + \frac{0.5}{3} + \frac{0.3}{4} + \frac{0.2}{5} \right\}$$

and

$$\underline{B} = \left\{ \frac{0.5}{2} + \frac{0.7}{3} + \frac{0.2}{4} + \frac{0.4}{5} \right\}$$

Find :

- (i) The set of loadings for which one expects that either material B or material D will be "safe".
- (ii) The set of loadings for which one expects that both material B and material D are "safe".
- (iii) The sets of loadings for which material D and material B are unsafe, respectively.
- (iv) The set of loadings for which the material is safe but the brittle material is in jeopardy.
- (v) The set of loadings for which the brittle material is safe but ductile material is in jeopardy.

OR

(b) Explain the following properties of fuzzy set : (CO2, CO3)

- (i) Law of excluded middle
- (ii) Law of contradiction
- (iii) Entropy of a fuzzy set

3. (a) Methane bio-filters can be used to oxidize methane using biological activities. It has become necessary to compare performance of two test columns, A and B. The methane outflow level at the surface, in non-dimensional units of $X = \{50, 100, 150, 200\}$, was detected and is tabulated below against the respective methane inflow into each test column. The following fuzzy sets represent the test columns : (CO2, CO3)

$$\tilde{A} = \left\{ \frac{0.15}{50} + \frac{0.25}{100} + \frac{0.5}{150} + \frac{0.7}{200} \right\} \text{ and } \tilde{B} = \left\{ \frac{0.2}{50} + \frac{0.3}{100} + \frac{0.6}{150} + \frac{0.65}{200} \right\}$$

Calculate the union, intersection, and the difference for the test columns.

OR

- (b) The membership function distribution of a fuzzy set is assumed to follow a Gaussian distribution with mean, $m = 200$ and standard deviation of 20. Determine 0.6-cut of this distribution. (CO2, CO3)
4. (a) We have a universe comprised of three elements, $X = \{a, b, c\}$, so the cardinal number is $n_x = 3$. Find its power set $P(X)$ (i.e, All possible sets of X constitute a special set called the power set, $P(X)$ and cardinality of the power set. (CO2, CO3)

OR

- (b) Explain the following : (CO2, CO3)
- (i) Complement of a fuzzy set
 - (ii) Intersection of fuzzy set
 - (iii) Union of fuzzy set
 - (iv) Algebraic sum of two fuzzy set
 - (v) Algebraic difference of two fuzzy set

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5. (a) Find :

(CO2, CO3)

$$A(x) = \{(x_1, 0.1)(x_2, 0.2)(x_3, 0.3)(x_4, 0.4)\}$$

and $B(x) = \{(x_1, 0.5)(x_2, 0.7)(x_3, 0.8)(x_4, 0.9)\}$

(i) $A(x).B(x)$

(ii) $4.A(x)$

(iii) $A^2(x)$

(iv) $A(x) + B(x)$

(v) $A(x) \oplus B(x)$

(vi) $A(x) - B(x)$

OR

(b) Suppose we have two fuzzy sets, $\tilde{A}$ defined on a universe of three discrete temperatures, $X = \{x_1, x_2, x_3\}$, and $\tilde{B}$ defined on a universe of two discrete pressures, $Y = \{y_1, y_2\}$, given as below : (CO2, CO3)

$$\tilde{A} = \frac{0.2}{x_1} + \frac{0.5}{x_2} + \frac{1}{x_3}$$

and

$$\tilde{B} = \frac{0.3}{y_1} + \frac{0.9}{y_2}$$

Find the fuzzy Cartesian product between them.

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B. TECH. (EE) (EIGHTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023
WIND AND SOLAR ENERGY SYSTEMS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What are advantages and disadvantages of Solar Collector ? Mention the applications of Solar Collector. (CO2)

OR

- (b) Discuss the principle of conversion of solar energy to heat energy conversion. (CO2)

2. (a) Discuss flat plate type solar collectors and concentrating focusing type solar collector. (CO3)

OR

- (b) Discuss working of conventional solar air heater. (CO3)

3. (a) How can global warming be controlled using solar energy ? (CO1)

OR

- (b) What is solar pond ? Discuss its principle of working with a neat diagram. (CO1)

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(2)

4. (a) Discuss the advantages and disadvantages of wind energy conversion system. (CO1)

OR

- (b) Discuss Betz coefficient in detail. Draw the block diagram for basic component of wind energy conversion system. (CO1)

5. (a) Explain briefly the factors that adversely affect the efficiency of a solar system. (CO2)

OR

- (b) What is photovoltaic effect ? Explain in brief. (CO2)

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B. TECH. (EE) (EIGHTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023
INTELLIGENT SENSORS AND INSTRUMENTATION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What do you understand by Instrumentation ? Use the suitable block diagram to explain instrumentation. (CO1)

OR

- (b) Explain in your words, the importance of the measurement in engineering. Give suitable examples to justify your answer. (CO1)

2. (a) Explain the difference between a sensor and transducer. Justify your answer with suitable examples. (CO1)

OR

- (b) Explain why MEMS stand for micro-electromechanical systems. Also, explain the working of MEMS with the help of an suitable example.

(CO1)

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(2)

3. (a) What do you understand by an actuator ? List the various types of actuators. Explain any *one* in brief. (CO1)

OR

- (b) What do you understand by a biosensor ? What are various applications of biosensors ? List the various types of biosensors. (CO1)
4. (a) What are various components of MEMS ? Briefly write their roles. Also, list the various biomedical applications of MEMS. (CO2)

OR

- (b) Differentiate between traditional sensors and modern, intelligent sensors. (CO2)
5. (a) What aspects of neuro-sensors do you comprehend ? What sets them apart from network sensors is the way they operate ? (CO2)

OR

- (b) Compare to traditional sensors, biosensors and neuro-sensors provide a variety of advantages that you should list. (CO2)